

AI IMAGE :

**Questions:****Q1. What is Artificial Intelligence?**

Artificial Intelligence, commonly known as AI, is a branch of computer science that focuses on creating systems that can perform tasks which usually require human intelligence. These tasks include learning from experience, understanding language, solving problems, recognizing patterns, and making decisions. AI works by using data and algorithms to identify patterns and provide useful results. It helps machines improve their performance over time. Today, AI is used in many areas such as healthcare, education, banking, online shopping, and entertainment. Examples include virtual assistants like Siri, recommendation systems on YouTube and Netflix, chatbots, and self-driving cars. AI is becoming an important technology because it helps people save time, increase efficiency, and solve complex problems.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI**Artificial Intelligence (AI)**

Artificial Intelligence is the broad concept of making machines capable of performing tasks that normally require human intelligence. AI includes problem solving, reasoning, learning, and understanding information.

Example: Face recognition in smartphones.

Machine Learning (ML)

Machine Learning is a subset of AI that allows systems to learn from data without being explicitly programmed for every task. Instead of following fixed rules, machine learning models improve their performance by analyzing patterns in data.

Example: Spam email detection.

Generative AI

Generative AI is a type of AI that creates new content such as text, images, music, videos, and code. It learns from large datasets and produces original outputs.

Example: ChatGPT generating text or AI image generators creating artwork.

Agentic AI

Agentic AI is advanced AI that can make decisions, plan tasks, and perform actions independently to achieve a goal. It can analyze situations, make choices, and take multiple steps without constant human guidance.

Example: AI systems that can schedule meetings or manage tasks automatically.

Q3. Explain any three open-source AI models available on Hugging Face

1. BERT

BERT stands for Bidirectional Encoder Representations from Transformers. It was developed by Google. BERT is designed to understand the meaning of words in context. It is used for text understanding tasks such as answering questions, sentiment analysis, and language translation. It is popular because it understands both left and right context of words.

2. GPT-2

GPT-2 is an advanced text generation model developed by OpenAI. It can generate human-like text based on a prompt. It is useful for writing assistance, summarization, chatbot systems, and content generation. GPT-2 became popular because it demonstrated strong natural language generation abilities.

3. Stable Diffusion

Stable Diffusion is an open-source AI model used for image generation. It creates images from text descriptions entered by the user. It is commonly used for digital art, design creation, and creative visual projects. Stable Diffusion is powerful because it allows users to create detailed images quickly.

Q4. What is the difference between traditional programming and AI-assisted development?

Traditional Programming

In traditional programming, the programmer writes detailed instructions for the computer to follow. The computer only performs tasks exactly as coded.

Example: If a programmer wants a calculator to add numbers, they must manually write the exact logic.

AI-Assisted Development

In AI-assisted development, AI tools help programmers by suggesting code, identifying mistakes, and automating repetitive tasks. AI can provide recommendations and improve coding speed.

Example: Tools like GitHub Copilot suggest code automatically while programming.

Main Difference

Traditional programming depends fully on manual coding, while AI-assisted development combines human skills with AI support to improve productivity.

Q5. Explain AI hallucinations with one example

AI hallucination happens when an AI system generates false, incorrect, or misleading information while sounding confident. This happens because AI predicts responses based on patterns in data rather than understanding truth like humans.

AI hallucinations can occur when:

- The AI lacks enough information
- The training data is incomplete
- The question is unclear

Example

If someone asks:

“Who invented Python?”

and the AI answers:

“Bill Gates invented Python in 1980.”

This is an AI hallucination because the information is false. The correct answer is that Python was created by **Guido van Rossum** and first released in 1991. This shows why AI-generated information should always be verified.